Astronomical Instrumentation Lab – IEE 3873

Construction of a small low resolution spectrometer for the visible

Purpose:

The purpose of the experience is to assemble a small spectrometer for the visible spectral range. Once ready the instrument must be characterized in the Lab. In particular the following parameters must be measured:

- Spectral range
- Spectral dispersion and resolution

The values obtained must be compared with the theoretical expectations.

Procedure:

- Use the mechanical and optical designs available to assemble the instrument,
- Possible missing parts must be purchased or manufactured,
- Compute the expected performances of the instrument using the information available and your knowledge of optics. Use first order approximation and Zemax, compare the results,
- Acquire spectra in the Lab and analyze them to measure the spectral dispersion and resolution, compare your results with the theoretical expectations,
- Write a complete report.